

Questions

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1. It is given that

$$y = x^2 e^{ax}$$

where a is a constant.

Prove by mathematical induction that, for all integers $n \geq 2$,

$$\frac{d^n y}{dx^n} = a^{n-2} (a^2 x^2 + 2nax + n(n-1)) e^{ax} \quad [6]$$

Solution

Let

$$P(n) : \frac{d^n y}{dx^n} = a^{n-2} (a^2 x^2 + 2nax + n(n-1)) e^{ax}$$

We prove $P(n)$ is true for all integers $n \geq 2$ by mathematical induction.

Base case: $n = 2$

Since $y = x^2 e^{ax}$,

$$\begin{aligned} \frac{dy}{dx} &= 2xe^{ax} + x^2(ae^{ax}) \\ &= (ax^2 + 2x)e^{ax} \end{aligned}$$

Differentiate again:

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} ((ax^2 + 2x)e^{ax}) \\ &= (2ax + 2)e^{ax} + (ax^2 + 2x)(ae^{ax}) \\ &= (2ax + 2 + a^2 x^2 + 2ax)e^{ax} \\ &= (a^2 x^2 + 4ax + 2)e^{ax} \end{aligned}$$

Now substitute $n = 2$ into the given formula:

$$\begin{aligned} a^{2-2} (a^2 x^2 + 2(2)ax + 2(2-1)) e^{ax} &= a^0 (a^2 x^2 + 4ax + 2) e^{ax} \\ &= (a^2 x^2 + 4ax + 2) e^{ax} \end{aligned}$$

This matches $\frac{d^2 y}{dx^2}$, so $P(2)$ is true.

Inductive Step

Now assume $P(k)$ is true for some integer $k \geq 2$. That is, assume

$$\frac{d^k y}{dx^k} = a^{k-2} (a^2 x^2 + 2kax + k(k-1)) e^{ax}$$

This is the induction hypothesis.

We must show that $P(k+1)$ is true.

Differentiate the induction hypothesis:

$$\frac{d^{k+1} y}{dx^{k+1}} = a^{k-2} \frac{d}{dx} [(a^2 x^2 + 2kax + k(k-1)) e^{ax}]$$

Using the product rule,

$$\begin{aligned} \frac{d^{k+1} y}{dx^{k+1}} &= a^{k-2} [(2a^2 x + 2ka) e^{ax} + (a^2 x^2 + 2kax + k(k-1)) a e^{ax}] \\ &= a^{k-2} [2a^2 x + 2ka + a^3 x^2 + 2ka^2 x + ak(k-1)] e^{ax} \end{aligned}$$

Now simplify the bracket:

$$\begin{aligned} 2a^2x + 2ka + a^3x^2 + 2ka^2x + ak(k-1) &= a^3x^2 + (2+2k)a^2x + (2k+k(k-1))a \\ &= a^3x^2 + 2(k+1)a^2x + k(k+1)a \end{aligned}$$

So

$$\begin{aligned} \frac{d^{k+1}y}{dx^{k+1}} &= a^{k-2} (a^3x^2 + 2(k+1)a^2x + k(k+1)a) e^{ax} \\ &= a^{k-1} (a^2x^2 + 2(k+1)ax + k(k+1)) e^{ax} \end{aligned}$$

Since

$$k(k+1) = (k+1)((k+1)-1)$$

this becomes

$$\frac{d^{k+1}y}{dx^{k+1}} = a^{(k+1)-2} (a^2x^2 + 2(k+1)ax + (k+1)((k+1)-1)) e^{ax}$$

This is exactly $P(k+1)$, so the inductive step is complete.

Conclusion:

Therefore, since $P(2)$ is true and $P(k) \Rightarrow P(k+1)$, it follows by mathematical induction that, for all integers $n \geq 2$,

$$\frac{d^ny}{dx^n} = a^{n-2} (a^2x^2 + 2nax + n(n-1)) e^{ax}$$

2. Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^n}{dx^n}(x^3 e^{-x}) = (-1)^n e^{-x} (x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2)) \quad [7]$$

Solution

We prove the result by mathematical induction.

Let

$$P(n) : \frac{d^n}{dx^n}(x^3 e^{-x}) = (-1)^n e^{-x} (x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2))$$

We must show that $P(n)$ is true for every positive integer n .

Base case: $n = 1$.

Differentiate $x^3 e^{-x}$ using the product rule:

$$\begin{aligned} \frac{d}{dx}(x^3 e^{-x}) &= 3x^2 e^{-x} + x^3(-e^{-x}) \\ &= e^{-x}(3x^2 - x^3) \\ &= (-1)e^{-x}(x^3 - 3x^2) \end{aligned}$$

Now substitute $n = 1$ into the required formula:

$$(-1)^1 e^{-x} (x^3 - 3(1)x^2 + 3(1)(0)x - (1)(0)(-1)) = (-1)e^{-x}(x^3 - 3x^2)$$

This matches the derivative found above, so $P(1)$ is true.

Inductive hypothesis.

Assume that $P(k)$ is true for some positive integer k .

That is, assume

$$\frac{d^k}{dx^k}(x^3 e^{-x}) = (-1)^k e^{-x} (x^3 - 3kx^2 + 3k(k-1)x - k(k-1)(k-2))$$

For convenience, let

$$P_k(x) = x^3 - 3kx^2 + 3k(k-1)x - k(k-1)(k-2)$$

Then the inductive hypothesis is

$$\frac{d^k}{dx^k}(x^3 e^{-x}) = (-1)^k e^{-x} P_k(x)$$

Inductive step.

We now show that $P(k+1)$ is true.

Differentiate the inductive hypothesis:

$$\begin{aligned} \frac{d^{k+1}}{dx^{k+1}}(x^3 e^{-x}) &= (-1)^k \frac{d}{dx}(e^{-x} P_k(x)) \\ &= (-1)^k ((-e^{-x})P_k(x) + e^{-x} P'_k(x)) \\ &= (-1)^k e^{-x} (P'_k(x) - P_k(x)) \\ &= (-1)^{k+1} e^{-x} (P_k(x) - P'_k(x)) \end{aligned}$$

Now find $P'_k(x)$:

$$P'_k(x) = 3x^2 - 6kx + 3k(k-1)$$

So

$$\begin{aligned} P_k(x) - P'_k(x) &= (x^3 - 3kx^2 + 3k(k-1)x - k(k-1)(k-2)) \\ &\quad - (3x^2 - 6kx + 3k(k-1)) \\ &= x^3 - 3(k+1)x^2 + (3k(k-1) + 6k)x \\ &\quad - (k(k-1)(k-2) + 3k(k-1)) \\ &= x^3 - 3(k+1)x^2 + 3k(k+1)x - k(k-1)(k+1) \\ &= x^3 - 3(k+1)x^2 + 3(k+1)kx - (k+1)k((k+1)-2) \end{aligned}$$

Therefore

$$\frac{d^{k+1}}{dx^{k+1}}(x^3 e^{-x}) = (-1)^{k+1} e^{-x} (x^3 - 3(k+1)x^2 + 3(k+1)kx - (k+1)k((k+1)-2))$$

This is exactly the required formula with $n = k + 1$, so $P(k + 1)$ is true.

Conclusion.

Since $P(1)$ is true, and $P(k) \Rightarrow P(k + 1)$, the statement holds for all positive integers n by mathematical induction.

Hence

$$\frac{d^n}{dx^n}(x^3 e^{-x}) = (-1)^n e^{-x} (x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2))$$

for every positive integer n .

3. Given that

$$y = e^{-2x} \cosh 3x$$

prove by induction that for $n \in \mathbb{N}$

$$\frac{d^n y}{dx^n} = e^{-2x} \left(\frac{1 + (-5)^n}{2} \cosh 3x + \frac{1 - (-5)^n}{2} \sinh 3x \right) \quad [6]$$

Solution

We prove the result by induction on n .

Let $P(n)$ be the statement

$$\frac{d^n y}{dx^n} = e^{-2x} \left(\frac{1 + (-5)^n}{2} \cosh 3x + \frac{1 - (-5)^n}{2} \sinh 3x \right)$$

where

$$y = e^{-2x} \cosh 3x$$

Base case: $n = 1$

Differentiate y using the product rule. Also,

$$\frac{d}{dx}(\cosh 3x) = 3 \sinh 3x$$

So

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(e^{-2x} \cosh 3x) \\ &= e^{-2x}(-2) \cosh 3x + e^{-2x}(3 \sinh 3x) \\ &= e^{-2x}(-2 \cosh 3x + 3 \sinh 3x) \end{aligned}$$

Now check this against the required formula when $n = 1$:

$$\frac{1 + (-5)^1}{2} = \frac{1 - 5}{2} = -2, \quad \frac{1 - (-5)^1}{2} = \frac{1 + 5}{2} = 3$$

So

$$\frac{dy}{dx} = e^{-2x} \left(\frac{1 + (-5)^1}{2} \cosh 3x + \frac{1 - (-5)^1}{2} \sinh 3x \right)$$

Hence $P(1)$ is true.

Inductive hypothesis

Assume that $P(n)$ is true for some $n \in \mathbb{N}$. That is, assume

$$\frac{d^n y}{dx^n} = e^{-2x} \left(\frac{1 + (-5)^n}{2} \cosh 3x + \frac{1 - (-5)^n}{2} \sinh 3x \right)$$

For simplicity, write

$$A = \frac{1 + (-5)^n}{2}, \quad B = \frac{1 - (-5)^n}{2}$$

Then the inductive hypothesis becomes

$$\frac{d^n y}{dx^n} = e^{-2x}(A \cosh 3x + B \sinh 3x)$$

Inductive step

We now differentiate to find $\frac{d^{n+1}y}{dx^{n+1}}$:

$$\begin{aligned} \frac{d^{n+1}y}{dx^{n+1}} &= \frac{d}{dx} [e^{-2x}(A \cosh 3x + B \sinh 3x)] \\ &= -2e^{-2x}(A \cosh 3x + B \sinh 3x) + e^{-2x}(3A \sinh 3x + 3B \cosh 3x) \\ &= e^{-2x} [(-2A + 3B) \cosh 3x + (3A - 2B) \sinh 3x] \end{aligned}$$

Now simplify the coefficients.

First,

$$\begin{aligned} -2A + 3B &= -2 \left(\frac{1 + (-5)^n}{2} \right) + 3 \left(\frac{1 - (-5)^n}{2} \right) \\ &= \frac{-2 - 2(-5)^n + 3 - 3(-5)^n}{2} \\ &= \frac{1 - 5(-5)^n}{2} \\ &= \frac{1 + (-5)^{n+1}}{2} \end{aligned}$$

since

$$(-5)^{n+1} = -5(-5)^n$$

Also,

$$\begin{aligned} 3A - 2B &= 3 \left(\frac{1 + (-5)^n}{2} \right) - 2 \left(\frac{1 - (-5)^n}{2} \right) \\ &= \frac{3 + 3(-5)^n - 2 + 2(-5)^n}{2} \\ &= \frac{1 + 5(-5)^n}{2} \\ &= \frac{1 - (-5)^{n+1}}{2} \end{aligned}$$

Therefore

$$\frac{d^{n+1}y}{dx^{n+1}} = e^{-2x} \left(\frac{1 + (-5)^{n+1}}{2} \cosh 3x + \frac{1 - (-5)^{n+1}}{2} \sinh 3x \right)$$

So $P(n+1)$ is true.

Since $P(1)$ is true, and $P(n) \Rightarrow P(n+1)$, the result follows by induction for all $n \in \mathbb{N}$.

Hence

$$\frac{d^n y}{dx^n} = e^{-2x} \left(\frac{1 + (-5)^n}{2} \cosh 3x + \frac{1 - (-5)^n}{2} \sinh 3x \right) \quad \text{for all } n \in \mathbb{N}$$

4. Let

$$y = x^2 \cosh x$$

Prove by induction that, for all integers $n \geq 1$,

$$\frac{d^{2n}y}{dx^{2n}} = x^2 \cosh x + 4nx \sinh x + 2n(2n-1) \cosh x$$

[6]

Solution

Let

$$P(n) : \frac{d^{2n}y}{dx^{2n}} = x^2 \cosh x + 4nx \sinh x + 2n(2n-1) \cosh x$$

We prove $P(n)$ by mathematical induction for all integers $n \geq 1$.

Base case: $n = 1$.

Since $y = x^2 \cosh x$,

$$\frac{dy}{dx} = 2x \cosh x + x^2 \sinh x$$

and so

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx}(2x \cosh x + x^2 \sinh x) \\ &= (2 \cosh x + 2x \sinh x) + (2x \sinh x + x^2 \cosh x) \\ &= x^2 \cosh x + 4x \sinh x + 2 \cosh x \end{aligned}$$

This is

$$x^2 \cosh x + 4(1)x \sinh x + 2 \cdot 1(2 \cdot 1 - 1) \cosh x$$

so $P(1)$ is true.

Inductive step.

Assume $P(k)$ is true for some integer $k \geq 1$. Then

$$\frac{d^{2k}y}{dx^{2k}} = x^2 \cosh x + 4kx \sinh x + 2k(2k-1) \cosh x$$

We must show that $P(k+1)$ is true.

Differentiate once:

$$\begin{aligned} \frac{d^{2k+1}y}{dx^{2k+1}} &= \frac{d}{dx}(x^2 \cosh x + 4kx \sinh x + 2k(2k-1) \cosh x) \\ &= (2x \cosh x + x^2 \sinh x) + 4k(\sinh x + x \cosh x) + 2k(2k-1) \sinh x \\ &= x^2 \sinh x + (4k+2)x \cosh x + (4k+2k(2k-1)) \sinh x \\ &= x^2 \sinh x + (4k+2)x \cosh x + 2k(2k+1) \sinh x \end{aligned}$$

Differentiate again:

$$\begin{aligned} \frac{d^{2k+2}y}{dx^{2k+2}} &= \frac{d}{dx}(x^2 \sinh x + (4k+2)x \cosh x + 2k(2k+1) \sinh x) \\ &= (2x \sinh x + x^2 \cosh x) + (4k+2)(\cosh x + x \sinh x) + 2k(2k+1) \cosh x \\ &= x^2 \cosh x + (2+(4k+2))x \sinh x + ((4k+2)+2k(2k+1)) \cosh x \\ &= x^2 \cosh x + 4(k+1)x \sinh x + (4k+2+4k^2+2k) \cosh x \\ &= x^2 \cosh x + 4(k+1)x \sinh x + (4k^2+6k+2) \cosh x \end{aligned}$$

Now

$$4k^2 + 6k + 2 = 2(k+1)(2k+1) = 2(k+1)(2(k+1)-1)$$

So

$$\frac{d^{2k+2}y}{dx^{2k+2}} = x^2 \cosh x + 4(k+1)x \sinh x + 2(k+1)(2(k+1)-1) \cosh x$$

This is exactly $P(k+1)$. Therefore $P(k) \Rightarrow P(k+1)$.

Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, it follows by induction that for all integers $n \geq 1$,

$$\frac{d^{2n}y}{dx^{2n}} = x^2 \cosh x + 4nx \sinh x + 2n(2n-1) \cosh x$$

5. A function is defined by $y = f(t)$ where $f(t) = (1 + at) \ln(1 + at)$ and a is a constant.

(a) By considering $\frac{d^2y}{dt^2}$, $\frac{d^3y}{dt^3}$, $\frac{d^4y}{dt^4}$ and $\frac{d^5y}{dt^5}$, make a conjecture for a general formula for $\frac{d^n y}{dt^n}$ in terms of n , a and t for any integer $n \geq 2$. [3]

(b) Use induction to prove the formula conjectured in part (a). [4]

(c) In the case when $f(t) = (1 + 3t) \ln(1 + 3t)$, find the rate at which the 6th derivative of $f(t)$ is varying when $t = \frac{5}{3}$. [2]

Solution

(a) Let

$$y = (1 + at) \ln(1 + at)$$

First differentiate once:

$$\begin{aligned} \frac{dy}{dt} &= \frac{d}{dt}(1 + at) \cdot \ln(1 + at) + (1 + at) \cdot \frac{d}{dt}(\ln(1 + at)) \\ &= a \ln(1 + at) + (1 + at) \cdot \frac{a}{1 + at} \\ &= a \ln(1 + at) + a \end{aligned}$$

Now find the next derivatives:

$$\begin{aligned} \frac{d^2y}{dt^2} &= \frac{d}{dt}(a \ln(1 + at) + a) = a \cdot \frac{a}{1 + at} = \frac{a^2}{1 + at} \\ \frac{d^3y}{dt^3} &= \frac{d}{dt}(a^2(1 + at)^{-1}) = a^2(-1)(1 + at)^{-2} \cdot a = -\frac{a^3}{(1 + at)^2} \\ \frac{d^4y}{dt^4} &= \frac{d}{dt}(-a^3(1 + at)^{-2}) = -a^3(-2)(1 + at)^{-3} \cdot a = \frac{2a^4}{(1 + at)^3} \\ \frac{d^5y}{dt^5} &= \frac{d}{dt}(2a^4(1 + at)^{-3}) = 2a^4(-3)(1 + at)^{-4} \cdot a = -\frac{6a^5}{(1 + at)^4} \end{aligned}$$

The pattern is:

$$\frac{d^2y}{dt^2} = \frac{a^2}{(1 + at)^1}, \quad \frac{d^3y}{dt^3} = -\frac{a^3}{(1 + at)^2}, \quad \frac{d^4y}{dt^4} = \frac{2a^4}{(1 + at)^3}, \quad \frac{d^5y}{dt^5} = -\frac{6a^5}{(1 + at)^4}$$

and the numbers 1, 1, 2, 6 are 0!, 1!, 2!, 3!.

So the conjecture is

$$f^{(n)}(t) = \frac{(-1)^n(n-2)!a^n}{(1 + at)^{n-1}} \quad \text{for all integers } n \geq 2$$

(b) We prove by induction that

$$f^{(n)}(t) = \frac{(-1)^n(n-2)!a^n}{(1 + at)^{n-1}} \quad (n \geq 2)$$

Base case: $n = 2$.

From part (a),

$$f''(t) = \frac{a^2}{1 + at}$$

The formula gives

$$\frac{(-1)^2(2-2)!a^2}{(1 + at)^{2-1}} = \frac{1 \cdot 0! a^2}{1 + at} = \frac{a^2}{1 + at}$$

so the result is true for $n = 2$.

Induction hypothesis.

Assume that for some integer $k \geq 2$,

$$f^{(k)}(t) = \frac{(-1)^k (k-2)! a^k}{(1+at)^{k-1}}$$

Induction step.

Differentiate both sides:

$$\begin{aligned} f^{(k+1)}(t) &= \frac{d}{dt} \left[(-1)^k (k-2)! a^k (1+at)^{-(k-1)} \right] \\ &= (-1)^k (k-2)! a^k \cdot (-(k-1)(1+at)^{-k} \cdot a) \\ &= (-1)^{k+1} (k-1)(k-2)! a^{k+1} (1+at)^{-k} \\ &= \frac{(-1)^{k+1} (k-1)! a^{k+1}}{(1+at)^k} \end{aligned}$$

This is exactly the required formula with $n = k + 1$, since

$$(k+1) - 2 = k - 1 \quad \text{and} \quad (k+1) - 1 = k$$

So, if the formula is true for $n = k$, it is also true for $n = k + 1$.

Since it is true for $n = 2$, it follows by induction that

$$f^{(n)}(t) = \frac{(-1)^n (n-2)! a^n}{(1+at)^{n-1}} \quad \text{for all integers } n \geq 2$$

(c) Here $a = 3$, so

$$f^{(n)}(t) = \frac{(-1)^n (n-2)! 3^n}{(1+3t)^{n-1}}$$

The rate at which the 6th derivative is varying is the 7th derivative:

$$f^{(7)}(t) = \frac{(-1)^7 (7-2)! 3^7}{(1+3t)^6} = -\frac{5! 3^7}{(1+3t)^6}$$

When $t = \frac{5}{3}$,

$$1 + 3t = 1 + 3 \cdot \frac{5}{3} = 6$$

Hence

$$\begin{aligned} f^{(7)}\left(\frac{5}{3}\right) &= -\frac{120 \cdot 3^7}{6^6} \\ &= -\frac{120 \cdot 3^7}{(2 \cdot 3)^6} \\ &= -\frac{120 \cdot 3}{2^6} \\ &= -\frac{360}{64} = -\frac{45}{8} \end{aligned}$$

Therefore, the rate is

$$-\frac{45}{8}$$

6. The function $f(x)$ is defined by $f(x) = (1+x)^{-\frac{1}{2}}$, for $x > -1$.

(a) Prove by mathematical induction that the n th derivative of $f(x)$, $f^{(n)}(x)$, for all $n \geq 1$, is given by

$$f^{(n)}(x) = \frac{(-1)^n (2n)!}{2^{2n} n! (1+x)^{n+\frac{1}{2}}} \quad [4]$$

(b) Hence prove that

$$(1+x)^{-\frac{1}{2}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \dots + (-1)^n \frac{(2n)!}{2^{2n} (n!)^2} x^n + \dots$$

for $-1 < x \leq 1$.

[3]

Solution

(a) We prove the formula by mathematical induction on n .

Base case: $n = 1$

$$f(x) = (1+x)^{-1/2}$$

Differentiating once,

$$f'(x) = -\frac{1}{2}(1+x)^{-3/2}$$

The given formula gives, when $n = 1$,

$$\frac{(-1)^1 (2)!}{2^2 \cdot 1! (1+x)^{1+\frac{1}{2}}} = \frac{-2}{4(1+x)^{3/2}} = -\frac{1}{2}(1+x)^{-3/2}$$

So the result is true for $n = 1$.

Induction hypothesis

Assume that for some $k \geq 1$,

$$f^{(k)}(x) = \frac{(-1)^k (2k)!}{2^{2k} k! (1+x)^{k+\frac{1}{2}}}$$

Induction step

Differentiate both sides:

$$\begin{aligned} f^{(k+1)}(x) &= \frac{(-1)^k (2k)!}{2^{2k} k!} \cdot \frac{d}{dx} \left((1+x)^{-k-\frac{1}{2}} \right) \\ &= \frac{(-1)^k (2k)!}{2^{2k} k!} \left(-k - \frac{1}{2} \right) (1+x)^{-k-\frac{3}{2}} \\ &= \frac{(-1)^{k+1} (2k)!}{2^{2k} k!} \left(k + \frac{1}{2} \right) (1+x)^{-k-\frac{3}{2}} \\ &= \frac{(-1)^{k+1} (2k)!}{2^{2k} k!} \cdot \frac{2k+1}{2} (1+x)^{-k-\frac{3}{2}} \\ &= \frac{(-1)^{k+1} (2k)! (2k+1)}{2^{2k+1} k!} (1+x)^{-k-\frac{3}{2}} \end{aligned}$$

Now simplify the coefficient:

$$\begin{aligned}
\frac{(2k)!(2k+1)}{2^{2k+1}k!} &= \frac{2(k+1)(2k+1)(2k)!}{2^{2k+2}(k+1)k!} \\
&= \frac{(2k+2)(2k+1)(2k)!}{2^{2k+2}(k+1)!} \\
&= \frac{(2k+2)!}{2^{2k+2}(k+1)!}
\end{aligned}$$

Hence

$$f^{(k+1)}(x) = \frac{(-1)^{k+1}(2k+2)!}{2^{2k+2}(k+1)!(1+x)^{k+\frac{3}{2}}}$$

which is exactly the required formula with $n = k + 1$.

Therefore, by mathematical induction,

$$f^{(n)}(x) = \frac{(-1)^n(2n)!}{2^{2n}n!(1+x)^{n+\frac{1}{2}}} \quad \text{for all } n \geq 1$$

(b) Using the Maclaurin series,

$$f(x) = f(0) + \sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

First,

$$f(0) = (1+0)^{-1/2} = 1$$

From part (a), for $n \geq 1$,

$$f^{(n)}(0) = \frac{(-1)^n(2n)!}{2^{2n}n!}$$

So

$$\begin{aligned}
(1+x)^{-1/2} &= 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \cdot \frac{(-1)^n(2n)!}{2^{2n}n!} x^n \\
&= 1 + \sum_{n=1}^{\infty} (-1)^n \frac{(2n)!}{2^{2n}(n!)^2} x^n
\end{aligned}$$

Hence

$$(1+x)^{-1/2} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \cdots + (-1)^n \frac{(2n)!}{2^{2n}(n!)^2} x^n + \cdots$$

7. It is given that

$$y = (ax + 1)^2 \ln(ax + 1)$$

where a is a positive constant. Prove by mathematical induction that, for every integer $n \geq 3$,

$$\frac{d^n y}{dx^n} = 2(-1)^{n-3} \frac{(n-3)!a^n}{(ax+1)^{n-2}}$$

[6]

Solution

Let

$$P(n) : \quad \frac{d^n y}{dx^n} = 2(-1)^{n-3} \frac{(n-3)!a^n}{(ax+1)^{n-2}}$$

We prove $P(n)$ is true for all integers $n \geq 3$ by mathematical induction.

Base case: $n = 3$.

Given

$$y = (ax + 1)^2 \ln(ax + 1)$$

Differentiate step by step.

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} ((ax+1)^2 \ln(ax+1) + (ax+1)^2 \frac{d}{dx} (\ln(ax+1))) \\ &= 2(ax+1) \cdot a \cdot \ln(ax+1) + (ax+1)^2 \cdot \frac{a}{ax+1} \\ &= 2a(ax+1) \ln(ax+1) + a(ax+1) \\ &= a(ax+1)(2 \ln(ax+1) + 1) \end{aligned}$$

Now differentiate again.

$$\begin{aligned} \frac{d^2 y}{dx^2} &= a \frac{d}{dx} [(ax+1)(2 \ln(ax+1) + 1)] \\ &= a \left[\frac{d}{dx} (ax+1)(2 \ln(ax+1) + 1) + (ax+1) \frac{d}{dx} (2 \ln(ax+1) + 1) \right] \\ &= a \left[a(2 \ln(ax+1) + 1) + (ax+1) \left(2 \cdot \frac{a}{ax+1} \right) \right] \\ &= a [a(2 \ln(ax+1) + 1) + 2a] \\ &= a^2 (2 \ln(ax+1) + 3) \end{aligned}$$

Differentiate a third time.

$$\begin{aligned} \frac{d^3 y}{dx^3} &= a^2 \frac{d}{dx} (2 \ln(ax+1) + 3) \\ &= a^2 \left(2 \cdot \frac{a}{ax+1} \right) \\ &= \frac{2a^3}{ax+1} \end{aligned}$$

Now substitute $n = 3$ into the claimed formula.

$$\begin{aligned} 2(-1)^{3-3} \frac{(3-3)!a^3}{(ax+1)^{3-2}} &= 2(-1)^0 \frac{0!a^3}{(ax+1)^1} \\ &= \frac{2a^3}{ax+1} \end{aligned}$$

This agrees with $\frac{d^3 y}{dx^3}$, so $P(3)$ is true.

Induction hypothesis.

Assume that for some integer $k \geq 3$,

$$\frac{d^k y}{dx^k} = 2(-1)^{k-3} \frac{(k-3)!a^k}{(ax+1)^{k-2}}$$

Inductive step.

We must show that this implies

$$\frac{d^{k+1} y}{dx^{k+1}} = 2(-1)^{k-2} \frac{(k-2)!a^{k+1}}{(ax+1)^{k-1}}$$

Starting from the induction hypothesis,

$$\begin{aligned} \frac{d^{k+1} y}{dx^{k+1}} &= \frac{d}{dx} \left(2(-1)^{k-3} (k-3)!a^k (ax+1)^{-(k-2)} \right) \\ &= 2(-1)^{k-3} (k-3)!a^k \frac{d}{dx} \left((ax+1)^{-(k-2)} \right) \\ &= 2(-1)^{k-3} (k-3)!a^k \left(-(k-2)(ax+1)^{-(k-1)} \cdot a \right) \\ &= 2 \left[-(-1)^{k-3} \right] (k-2)(k-3)!a^{k+1} (ax+1)^{-(k-1)} \\ &= 2(-1)^{k-2} (k-2)!a^{k+1} (ax+1)^{-(k-1)} \\ &= 2(-1)^{k-2} \frac{(k-2)!a^{k+1}}{(ax+1)^{k-1}} \end{aligned}$$

This is exactly the required form for $P(k+1)$.

So, assuming $P(k)$ is true has allowed us to prove $P(k+1)$ is true.

Conclusion.

Since $P(3)$ is true, and $P(k) \Rightarrow P(k+1)$ for all integers $k \geq 3$, it follows by mathematical induction that

$$\frac{d^n y}{dx^n} = 2(-1)^{n-3} \frac{(n-3)!a^n}{(ax+1)^{n-2}} \quad \text{for all integers } n \geq 3$$

Hence the result is proved.

8. The function f is such that $f''(x) = -f(x)$.

Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^{2n}}{dx^{2n}} (x^2 f(x)) = (-1)^n (x^2 f(x) - 4nx f'(x) - 2n(2n-1)f(x)) \quad [7]$$

Solution

We prove the result by mathematical induction on n .

Let $P(n)$ be the statement

$$\frac{d^{2n}}{dx^{2n}} (x^2 f(x)) = (-1)^n (x^2 f(x) - 4nx f'(x) - 2n(2n-1)f(x))$$

Base case: $n = 1$

We calculate the second derivative directly:

$$\begin{aligned} \frac{d}{dx} (x^2 f(x)) &= 2xf(x) + x^2 f'(x) \\ \frac{d^2}{dx^2} (x^2 f(x)) &= \frac{d}{dx} (2xf + x^2 f') \\ &= 2f + 2xf' + 2xf' + x^2 f'' \\ &= 2f + 4xf' + x^2 f'' \end{aligned}$$

Given that $f''(x) = -f(x)$, this becomes

$$\begin{aligned} \frac{d^2}{dx^2} (x^2 f(x)) &= 2f + 4xf' - x^2 f \\ &= -(x^2 f - 4xf' - 2f) \\ &= (-1)^1 (x^2 f - 4(1)xf' - 2(1)(2 \cdot 1 - 1)f) \end{aligned}$$

So $P(1)$ is true.

Inductive hypothesis

Assume that $P(k)$ is true for some positive integer k . Then

$$\frac{d^{2k}}{dx^{2k}} (x^2 f) = (-1)^k (x^2 f - 4kx f' - 2k(2k-1)f)$$

Inductive step

We must show that $P(k+1)$ is true.

Starting from the inductive hypothesis and differentiating twice,

$$\begin{aligned} \frac{d^{2k+2}}{dx^{2k+2}} (x^2 f) &= \frac{d^2}{dx^2} \left[\frac{d^{2k}}{dx^{2k}} (x^2 f) \right] \\ &= (-1)^k \frac{d^2}{dx^2} (x^2 f - 4kx f' - 2k(2k-1)f) \\ &= (-1)^k ((x^2 f)'' - 4k(xf')'' - 2k(2k-1)f'') \end{aligned}$$

Now work out each of these second derivatives.

First, from the base case calculation,

$$(x^2 f)'' = -x^2 f + 4xf' + 2f$$

Next,

$$\begin{aligned} (xf')' &= f' + xf'' \\ (xf')'' &= f'' + f'' + xf''' \\ &= 2f'' + xf''' \end{aligned}$$

Since $f'' = -f$, differentiating gives $f''' = -f'$. Therefore

$$(xf')'' = 2(-f) + x(-f') = -2f - xf'$$

so

$$(xf')'' = -xf' - 2f$$

Also,

$$f'' = -f$$

Substituting these into the inductive step,

$$\begin{aligned}\frac{d^{2k+2}}{dx^{2k+2}}(x^2f) &= (-1)^k [(-x^2f + 4xf' + 2f) - 4k(-xf' - 2f) - 2k(2k-1)(-f)] \\ &= (-1)^k [-x^2f + 4xf' + 2f + 4kxf' + 8kf + 2k(2k-1)f] \\ &= (-1)^k [-x^2f + 4(k+1)xf' + (2+8k+2k(2k-1))f]\end{aligned}$$

Now simplify the coefficient of f :

$$\begin{aligned}2 + 8k + 2k(2k-1) &= 2 + 8k + 4k^2 - 2k \\ &= 4k^2 + 6k + 2 \\ &= 2(k+1)(2k+1)\end{aligned}$$

Hence

$$\begin{aligned}\frac{d^{2k+2}}{dx^{2k+2}}(x^2f) &= (-1)^k [-x^2f + 4(k+1)xf' + 2(k+1)(2k+1)f] \\ &= (-1)^{k+1} [x^2f - 4(k+1)xf' - 2(k+1)(2k+1)f]\end{aligned}$$

Since $2(k+1) - 1 = 2k + 1$, this is exactly

$$\frac{d^{2(k+1)}}{dx^{2(k+1)}}(x^2f) = (-1)^{k+1} (x^2f - 4(k+1)xf' - 2(k+1)(2(k+1)-1)f)$$

So $P(k+1)$ is true.

Conclusion

$P(1)$ is true, and $P(k) \Rightarrow P(k+1)$. Therefore, by mathematical induction,

$$\frac{d^{2n}}{dx^{2n}}(x^2f(x)) = (-1)^n (x^2f(x) - 4nxf'(x) - 2n(2n-1)f(x))$$

for every positive integer n .